



University of Colombo, Sri Lanka



*University of Colombo School of Computing*  
**BACHELOR OF SCIENCE IN COMPUTER SCIENCE**

First Year Examination - Semester II - UCSC AY22 [held in March 2026]

**SCS1308 — Foundations of Algorithms**

(Two (2) Hours)

**Answer ALL questions**

Number of Pages = 13

Number of Questions = 4

To be completed by the candidate

Index Number:

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**Important Instructions to Candidates:**

- I. Students should answer in English **only** using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number **CLEARLY** on every page of this Question paper.
- IV. Answer **ALL** questions. This paper consists of Four (4) questions on Thirteen (13) pages (including the Cover Page). All questions carry equal marks.
- V. Programmable Calculators and any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches and mobile phones, are **not** allowed.
- VI. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate.

**To be completed by the examiners**

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**Question 1(25 marks)**

(a). Consider the following Parallel AI Training Aggregation model.

A distributed AI system trains a model using 'n' data samples. The system works as follows:

- Data is divided into 3 equal subsets.
- Each subset is trained independently (recursively).
- The model parameters are merged, which requires time proportional to n.

(i). Write the recurrence relation for the above scenario, considering both recursive and non-recursive terms. Express your final answer in terms of  $T(n)$ .

[5 Marks]

**ANSWER IN THIS BOX**

(ii) Solve the recurrence equation using recursion tree method and provide complexity. Note that tree structure including root, depth and how leaves in some level's formation required to obtain full marks.

[ 10 marks]

**ANSWER IN THIS BOX**



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- (b) A logistics company wants to find the maximum delivery time among all shipments to detect delays.

```
Algorithm MAX_DELIVERY_TIME(T[0...n-1])
  maxTime ← T[0]
  for i ← 1 to n-1 do
    if T[i] > maxTime then
      maxTime ← T[i]
  return maxTime
```

Where:

T[i] = delivery time of shipment i

- (i) What is the loop invariant of the above code ?

[2 marks]

**ANSWER IN THIS BOX**

- (ii) Prove the correctness of the loop invariant for initialization, maintenance and termination phases.

[8 marks]

**ANSWER IN THIS BOX**

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**Question 2 (25 marks)**

(a) Consider vertex cover problem.

A vertex cover of an undirected graph is a set of vertices  $V'$  such that for every edge  $(u,v)$ , either  $u$  or  $v$  or both are in  $V'$ . The problem is to find a cover of minimum size.

VERTEX-COVER Problem

Input: A graph  $G$  and a number  $k$ .

Output: YES iff  $G$  has a vertex cover of size  $k$

Show VERTEX-COVER is NP Complete. You may assume Clique problem is NP Complete.

[15 Marks]

**ANSWER IN THIS BOX**

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(b) Consider the details of the following E-commerce platform which is an online store tracks product IDs.

- Hash function:  $h(k) = k \bmod m$
- Collision resolution: using Quadratic probing  $H_i(k) = (h(k) + i^2) \bmod m$
- Initial size:  $m = 11$
- Rehash when load factor  $\geq 0.75$
- New size ( $m$ ) = next prime after doubling

Insert following IDs to the hash table : 21,32,43,14,25,36,47,58

(i) What is the final hash table size ?

[2 marks]

**ANSWER IN THIS BOX**

(ii) Design a new hash table and demonstrate the placement of elements after applying the hash function by filling the following table with appropriate slot.

[8 marks]

**ANSWER IN THIS BOX**

| Key | Slot ( $m = \dots\dots\dots$ ) |
|-----|--------------------------------|
| 21  |                                |
| 32  |                                |
| 43  |                                |
| 14  |                                |
| 25  |                                |
| 36  |                                |
| 47  |                                |
| 58  |                                |



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**Question 3 (25 marks)**

- (a) (i) Insert numbers 10,20,30,40,50,60,70,80 sequentially into an AVL tree. Provide **only** the final balanced AVL Tree.

[4 marks]

**ANSWER IN THIS BOX**

- (ii) Build a Red-Black tree by inserting the given nodes into an initially empty tree, indicating all balancing operations.

50,30,70, 20,40,60,80,10

[8 marks]

**ANSWER IN THIS BOX**

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(b) Delete the root node and illustrate the resulting balanced Red-Black tree, clearly showing all recoloring and rotation steps.

[4 Marks]

**ANSWER IN THIS BOX**

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(c) A hospital assigns 3-digit priority codes to emergency patients. Lower numbers indicate higher urgency. Sort the following patient codes in ascending order using an appropriate sorting mechanism.

329,145,278,093,410,256

- (i) Select an appropriate sorting algorithm for processing  $n$  patients and justify your choice by stating its time complexity.

[1 Marks]

**ANSWER IN THIS BOX**

- (ii) Determine the final sorted sequence in ascending order and clearly demonstrate each step of the sorting procedure.

[5 Marks]

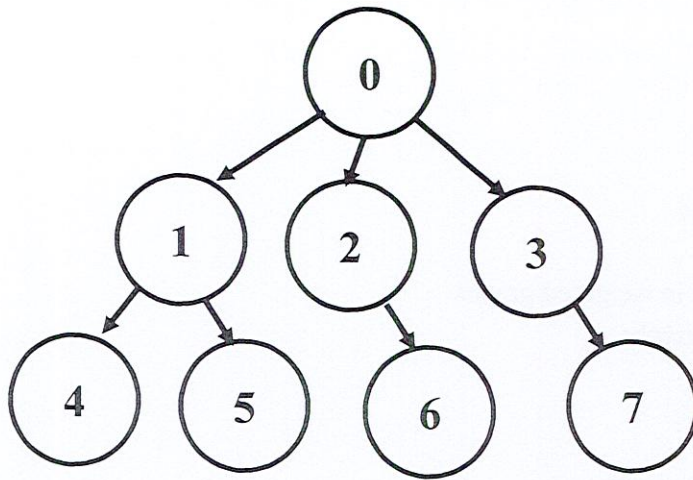
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(d) Consider the following tree data structure. Provide the BFS (Breadth-First Search) and DFS (Depth-First Search) traversal orders of the tree.



[3 Marks]

**ANSWER IN THIS BOX**

**Question 4 ( 25 marks)**

Select the most **suitable answer** for each of the following MCQ questions and mark your choice in the provided boxes.

| Question | Your choice<br>(a,b,c,d,e) |
|----------|----------------------------|
| 1        |                            |
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1. In a graph with 1,000 vertices and 10,000 edges, which statement is TRUE?

- A) DFS is  $O(1000^2)$
- B) BFS is  $O(1000 + 10000)$
- C) DFS is  $O(10000^2)$
- D) BFS is  $O(\log V)$
- E) DFS is  $O(VE)$

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2. Which data structure pairing is correct?

- A) BFS → Stack, DFS → Queue
- B) BFS → Queue, DFS → Stack
- C) BFS → Heap, DFS → Queue
- D) BFS → Recursion only
- E) DFS → Queue only

3. Which statement is TRUE for BFS?

- A) It uses recursion exclusively
- B) The path found is always minimal in weighted graphs
- C) It finds shortest path in weighted graphs
- D) It may not visit all vertices
- E) It cannot handle cycles

4. Which property ensures Red-Black trees are balanced?

- A) Height  $\leq 2 \log n$
- B) All left children < node
- C) Leaf nodes are black
- D) No consecutive red nodes + black-height property
- E) In-order traversal sorted

5. After DSW algorithm completes, the resulting BST is:

- A) Completely balanced
- B) Skewed
- C) Red-Black tree
- D) Heap
- E) AVL tree

6. The in-order traversal of a BST produces:

- A) Pre-order sequence
- B) Post-order sequence
- C) Sorted order of keys
- D) Reverse sorted order
- E) Random order

7. Perfect hashing is most suitable when:

- A) Dynamic set
- B) Frequent insertions
- C) Static key set
- D) Large load factor
- E) Unknown keys

8. Red-Black trees guarantee no path is more than twice the length of another because:

- A) Height rule
- B) Black-height property
- C) Degree rule
- D) BST property
- E) Heap property



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9. Radix sort becomes inefficient when:

- A) Keys large
- B) Digit count large
- C) Data sorted
- D) Using counting sort
- E) Memory small

10. If A reduces to B in polynomial time and A is NP-Complete, then B is:

- A) NP
- B) NP-Hard
- C) P
- D) P SPACE
- E) Exponential time

11. Which statement is TRUE?

- A) All NP-Hard problems are NP
- B) All NP-Complete problems are NP-Hard
- C) All NP problems are NP-Complete
- D)  $P \subset \text{NP-Hard}$
- E)  $\text{NP-Hard} \subset \text{NP}$

12. If a polynomial-time algorithm exists for 3-SAT, which is true?

- A) Only SAT becomes polynomial
- B) Only NP-hard problems solved
- C)  $P = \text{NP}$
- D)  $\text{PSPACE} = \text{NP}$
- E) Nothing changes

13.  $T(n) = 2T(n/2) + n / \log n$ ; Which is correct?

- A)  $\Theta(n)$
- B)  $\Theta(n \log n)$
- C)  $\Theta(n \log \log n)$
- D)  $\Theta(n^2)$
- E)  $\Theta(\log n)$

14. Height of a skewed BST with n nodes:

- A)  $\log n$
- B) n
- C)  $\sqrt{n}$
- D)  $n \log n$
- E) Constant

15. Interpolation search works best when:

- A) Data random
- B) Data unsorted
- C) Uniformly distributed
- D) Reverse sorted
- E) Duplicate values

16. Shell sort improves insertion sort by:

- A) Dividing array
- B) Using gaps
- C) Using recursion

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- D) Hashing
- E) Tree balancing

17. Handling NP-Hard problems includes:

- A) Exact polynomial solutions
- B) Approximation algorithms
- C) Binary search
- D) Hashing
- E) Recursion trees

18. A problem is NP if:

- A) It is unsolvable
- B) Solution verifiable in polynomial time
- C) Requires exponential time
- D) It is optimization problem
- E) It is recursive

19. What is the time complexity of the recursive function?

$$T(n) = T(n - 1) + 5$$

- A)  $O(\log n)$
- B)  $O(n)$
- C)  $O(n \log n)$
- D)  $O(n^2)$
- E)  $O(2^n)$

20. After creating the vine, DSW performs compress operations. The purpose is to:

- A) Convert to AVL
- B) Create perfect complete BST
- C) Reduce tree height by left rotations
- D) Convert to heap
- E) Detect imbalance

21. Given the expression:

$$a - b - c - d$$

Using left associativity, the root of the expression tree is:

- A) a
- B) b
- C) c
- D) First '-'
- E) Last '-'

22. For  $n = 15$  nodes, after DSW balancing, the resulting tree height is approximately:

- A) 15
- B) 8
- C) 3
- D) 4
- E) 2

23. The DSW algorithm first converts a BST with  $n$  nodes into a vine. What is the number of right rotations required in worst case?

- A)  $\log n$
- B)  $n - 1$
- C)  $n \log n$
- D)  $n^2$
- E)  $\sqrt{n}$



24. Suppose a program  $\text{HALT}(P, I)$  correctly determines whether program  $P$  halts on input  $I$ . What happens if we construct a program  $D$  that runs:

If  $\text{HALT}(P, P) = \text{TRUE} \rightarrow \text{loop forever}$   
 Else halt immediately  
 and then execute  $\text{HALT}(D, D)$ ?

- A)  $D$  halts
- B)  $D$  loops forever
- C) Contradiction arises
- D)  $\text{HALT}$  works correctly
- E)  $D$  becomes NP-Complete

25. Which statement is TRUE about the Halting Problem?

- A) It is NP-Complete
- B) It is NP-Hard
- C) It is undecidable
- D) It is solvable in exponential time
- E) It is PSPACE-Complete

\*\*\*\*\*

Here are the red-black deletion cases.

| Case # | Check condition                                                                                                                            | Action                                                                                                                                                                                                                                     |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | If node to be delete is a red leaf node                                                                                                    | Just remove it from the tree                                                                                                                                                                                                               |
| 2      | If DB node is root                                                                                                                         | Remove the DB and root node becomes black.                                                                                                                                                                                                 |
| 3      | (a) If DB's sibling is black, and<br>(b) DB's sibling's children are black                                                                 | (a) Remove the DB (if null DB then delete the node and for other nodes remove the DB sign)<br>(b) Make DB's sibling red.<br>(c) If DB's parent is black, make it DB, else make it black                                                    |
| 4      | If DB's sibling is red                                                                                                                     | (a) Swap color DB's parent with DB's sibling<br>(b) Perform rotation at parent node in the direction of DB node<br>(c) Check which case can be applied to this new tree and perform that action                                            |
| 5      | (a) DB's sibling is black<br>(b) DB's sibling's child which is far from DB is black<br>(c) DB's sibling's child which is near to DB is red | (a) Swap color of sibling with sibling's red child<br>(b) Perform rotation at sibling node in direction opposite of DB node<br>(c) Apply case 6                                                                                            |
| 6      | (a) DB's sibling is black, and<br>(b) DB's sibling's far child is red (remember this node)                                                 | (a) Swap color of DB's parent with DB's sibling's color<br>(b) Perform rotation at DB's parent in direction of DB<br>(c) Remove DB sign and make the node normal black node<br>(d) Change colour of DB's sibling's far red child to black. |